Arya Pramod

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.77
Amrita Vidyalayam Puthiyakavu	Class XII (CBSE)	2019	93.80%
Amrita Vidyalayam Puthiyakavu	Class X (CBSE)	2017	10.00%

SKILLS

Circuit Design Tools : Cadence Virtuoso, LTSpice, Vivado

Programming : Verilog, Python, C++

RELEVANT COURSES

Analog Domain : Analog IC Design, Analog IC Lab

Digital Domain : Introduction to VLSI Design, Digital IC Design, Digital IC Design Lab

Memory Domain : Memory Circuits and System Design

Device Domain : Semiconductor Device Modelling, VLSI technology

PROJECTS

Low-Power Dynamic Comparator Design for 8-bit SAR ADC *(Sept 2026)*

- Designed a low-power dynamic comparator for an 8-bit SAR ADC and implemented the circuit at transistor level.
- Characterized the comparator for power consumption, decision time and input-referred offset across relevant operating conditions.

Designed Two Stage Miller Compensated Op-Amp using TSMC 180nm Technology Node

- Designed a Two-Stage Miller Compensated Operational Amplifier using TSMC 180nm CMOS technology, achieving CMRR = 76.27 dB, PM = 75.04°, SR = 5.18 V/s, GBW = 4.848 MHz at VDD = 1.8 V with a 5 pF load.
- Used Miller compensation capacitor with a series resistor to shift RHP zero to left-half plane, improving phase margin and ensuring frequency stability.

Design of 8-bit Asynchronous SAR ADC for 10 MSPS Operation *(Sept 2026 - Present)*

- Designing an 8-bit asynchronous SAR ADC targeting 10 MSPS operation, with ideal behavioral modeling and initial functional verification completed.
- Currently transitioning the design to practical transistor-level implementation and developing the key circuit blocks of the ADC.

WORK EXPERIENCE

MMRFIC Technology Pvt. Ltd. - Intern *(Sept 26 - Present)*

- Working on the design of an 8-bit asynchronous SAR ADC targeting 10 MSPS, with ideal behavioral modeling completed and practical transistor-level implementation in progress.
- Developing the comparator and binary-weighted CDAC using monotonic switching as part of the ongoing ADC implementation.

POSITIONS OF RESPONSIBILITY

- Appointed as the Teaching Assistant (TA) for the "Digital Circuits" course for the 2nd year UG students and managed key logistics of the course. Instructor : Dr. Asudeb Dutta
- Appointed as the Teaching Assistant (TA) for the VLSI Chip Design Bootcamp conducted by IIT Hyderabad in collaboration with TCS iON